高擎
HIGH TORQUE

JOINT MODULE PRODUCT MANUAL

SERVO JOINT MODULE
OF HIGH TORQUE

关节模组产品手册

高擎动力伺服关节模组

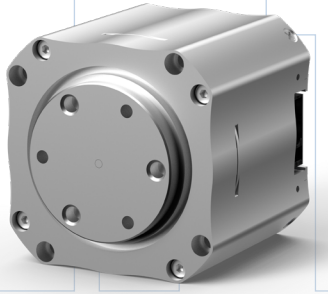


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修订日期：2026.06.02

高擎 HIGH TORQUE

High Torque



(HTDW-4530-02-DNE)

HTDW-4438-30-NE

行星关节模组

产品参数 Product Parameters

基本参数 Basic parameters

长宽 length and width (mm)	44
高度 Height (mm)	43.4
重量 weight (g)	237

减速机特性参数 Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

模组性能参数 Module Performance Parameters

减速比 Reduction ratio	30	堵转扭矩 Locked-rotor torque(Nm)	10
额定电压 Rated Voltage (V)	24	额定转速 Rated RPM(RPM)	40
额定电流 Rated Current (A)	0.7	空载转速 No-load RPM(RPM)	160
额定扭矩 Rated torque(Nm)	2	额定输出功率 Rated Output Power(W)	8.5

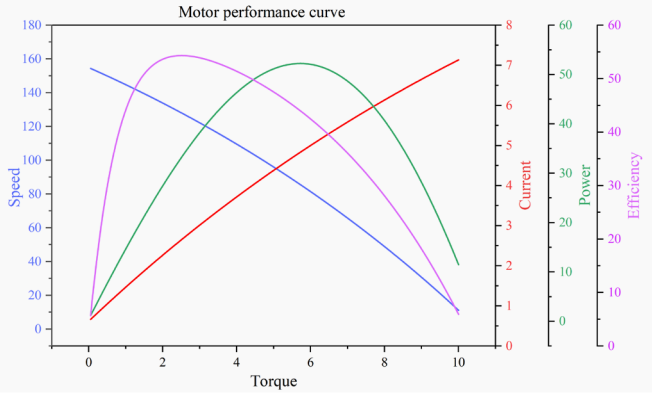
模组电气性能参数 Electrical Performance Parameters of the Module

电压范围 Voltage Range(V)	12-50.4	通讯协议 Communication	CANFD/CAN
峰值电流 Peak current(A)	10	扭矩控制精度 control accuracy(Nm/A)	±10%
转速控制精度 Speed control accuracy(Nm/A)	±10%	响应时间 Response time(us)	140
输入端编码器分辨率 High-speedend encoder resolution(bit)	14	输出端编码器分辨率 Low-speedend encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5	控制频率 Control frequency(Hz)	3K

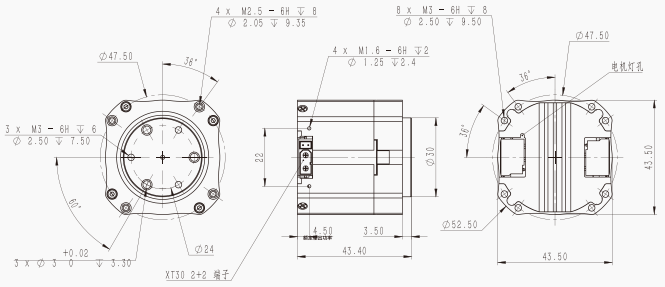
电机特性参数 Motor parameters

扭矩常数 Torque constant(Nm/A)	0.272	极对数 Polar logarithm	14

TN曲线 TN curve

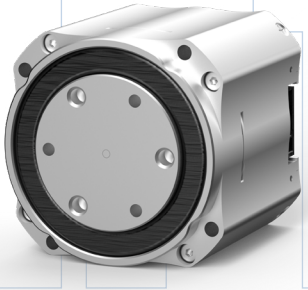


产品安装尺寸图纸 Product Installation Dimension Drawing



高擎 HIGH TORQUE

High Torque



(HTDW-4530-02-CNE)

HTDW-4438-30-NE-JC

(交叉滚子轴承版本) 行星关节模组

产品参数 Product Parameters

基本参数 Basic parameters

长宽 length and width (mm)	44
高度 Height (mm)	44.9
重量 weight (g)	237

减速机特性参数 Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

模组性能参数 Module Performance Parameters

减速比 Reduction ratio	30	堵转扭矩 Locked-rotor torque(Nm)	10
额定电压 Rated Voltage (V)	24	额定转速 Rated RPM(RPM)	40
额定电流 Rated Current (A)	0.7	空载转速 No-load RPM(RPM)	160
额定扭矩 Rated torque(Nm)	2	额定输出功率 Rated Output Power(W)	8.5

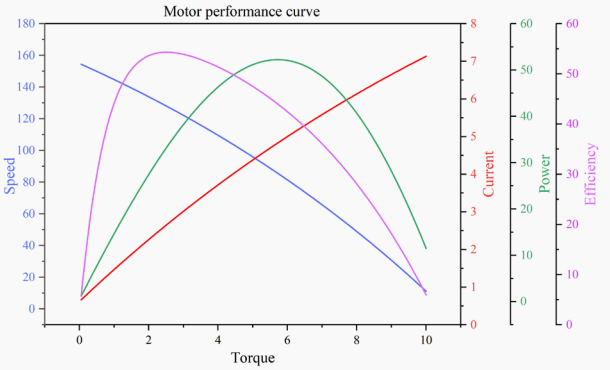
模组电气性能参数 Electrical Performance Parameters of the Module

电压范围 Voltage Range(V)	12-50.4	通讯协议 Communication	CANFD/CAN
峰值电流 Peak current(A)	10	扭矩控制精度 control accuracy(Nm/A)	±10%
转速控制精度 Speed control accuracy(Nm/A)	±10%	响应时间 Response time(us)	140
输入端编码器分辨率 High-speedend encoder resolution(bit)	14	输出端编码器分辨率 Low-speedend encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5	控制频率 Control frequency(Hz)	3K

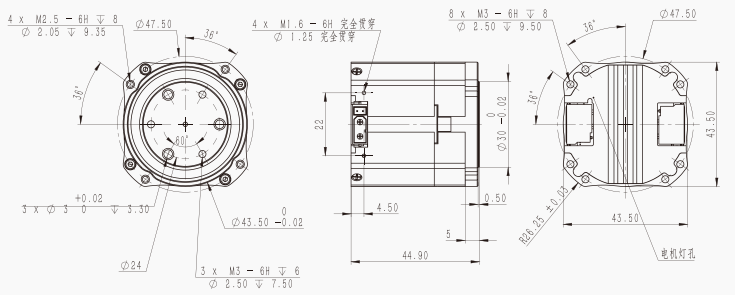
电机特性参数 Motor parameters

扭矩常数 Torque constant(Nm/A)	0.272	极对数 Polar logarithm	14

TN曲线 TN curve

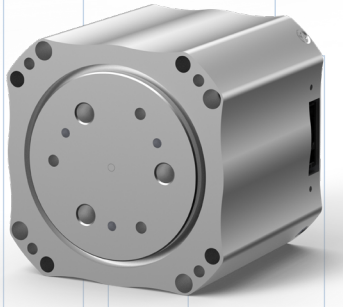


产品安装尺寸图纸 Product Installation Dimension Drawing



高擎 HIGH TORQUE

High Torque



HTDW-5022-02-DNE

行星关节模组

产品参数 Product Parameters

基本参数 Basic parameters

长宽 length and width (mm)	50
高度 Height (mm)	47.4
重量 weight (g)	322

减速机特性参数 Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

模组性能参数 Module Performance Parameters

减速比 Reduction ratio	22	堵转扭矩 Locked-rotor torque(Nm)	13
额定电压 Rated Voltage (V)	24	额定转速 Rated RPM(RPM)	82
额定电流 Rated Current (A)	2	空载转速 No-load RPM(RPM)	123
额定扭矩 Rated torque(Nm)	3.5	额定输出功率 Rated Output Power(W)	30

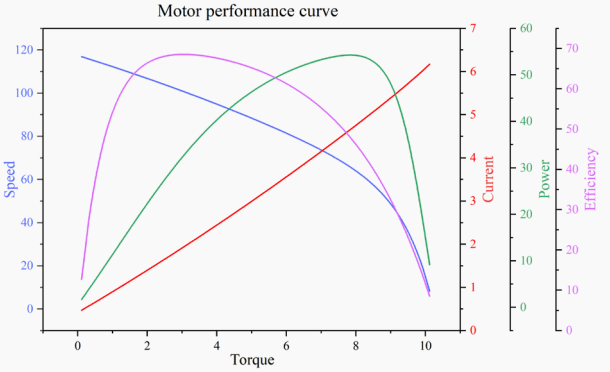
模组电气性能参数 Electrical Performance Parameters of the Module

电压范围 Voltage Range(V)	12-50.4	通讯协议 Communication	CANFD/CAN
峰值电流 Peak current(A)	10	扭矩控制精度 control accuracy(Nm/A)	±10%
转速控制精度 Speed control accuracy(Nm/A)	±10%	响应时间 Response time(us)	140
输入端编码器分辨率 High-speedend encoder resolution(bit)	14	输出端编码器分辨率 Low-speedend encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5	控制频率 Control frequency(Hz)	3K

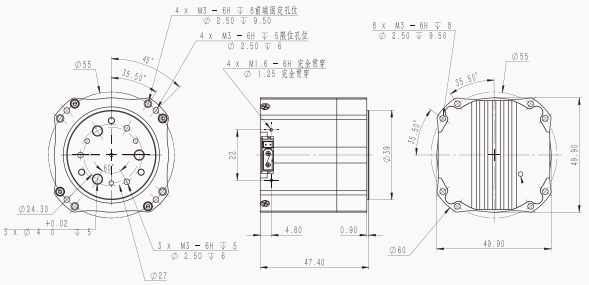
电机特性参数 Motor parameters

扭矩常数 Torque constant(Nm/A)	0.54	极对数 Polar logarithm	14

TN曲线 TN curve

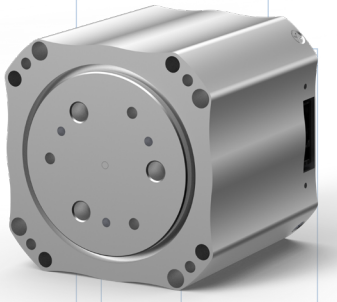


产品安装尺寸图纸 Product Installation Dimension Drawing



高擎 HIGH TORQUE

High Torque



HTDW-5036-02-DNE

行星关节模组

产品参数 Product Parameters

基本参数 Basic parameters

长宽 length and width (mm)	50
高度 Height (mm)	47.4
重量 weight (g)	323

减速机特性参数 Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

模组性能参数 Module Performance Parameters

减速比 Reduction ratio	36	堵转扭矩 Locked-rotor torque(Nm)	21
额定电压 Rated Voltage (V)	24	额定转速 Rated RPM(RPM)	50
额定电流 Rated Current (A)	2	空载转速 No-load RPM(RPM)	75
额定扭矩 Rated torque(Nm)	6	额定输出功率 Rated Output Power(W)	30

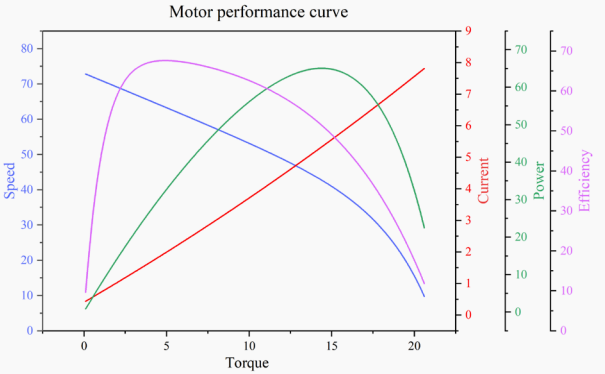
模组电气性能参数 Electrical Performance Parameters of the Module

电压范围 Voltage Range(V)	12-50.4	通讯协议 Communication	CANFD/CAN
峰值电流 Peak current(A)	10	扭矩控制精度 control accuracy(Nm/A)	±10%
转速控制精度 Speed control accuracy(Nm/A)	±10%	响应时间 Response time(us)	140
输入端编码器分辨率 High-speedend encoder resolution(bit)	14	输出端编码器分辨率 Low-speedend encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5	控制频率 Control frequency(Hz)	3K

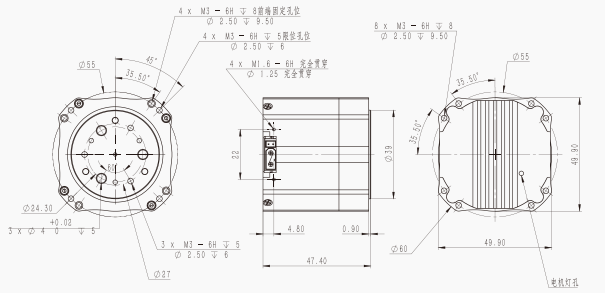
电机特性参数 Motor parameters

扭矩常数 Torque constant(Nm/A)	0.54	极对数 Polar logarithm	14

TN曲线 TN curve

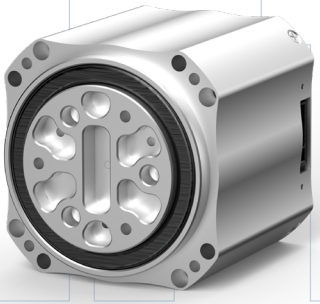


产品安装尺寸图纸 Product Installation Dimension Drawing



高擎 HIGH TORQUE

High Torque



HTDW-5036-02-CNE

(交叉滚子轴承版本) 行星关节模组

产品参数 Product Parameters

基本参数 Basic parameters

长宽 length and width (mm)	50
高度 Height (mm)	47.4
重量 weight (g)	345

减速机特性参数 Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

模组性能参数 Module Performance Parameters

减速比 Reduction ratio	36	堵转扭矩 Locked-rotor torque(Nm)	21
额定电压 Rated Voltage (V)	24	额定转速 Rated RPM(RPM)	50
额定电流 Rated Current (A)	2	空载转速 No-load RPM(RPM)	75
额定扭矩 Rated torque(Nm)	6	额定输出功率 Rated Output Power(W)	30

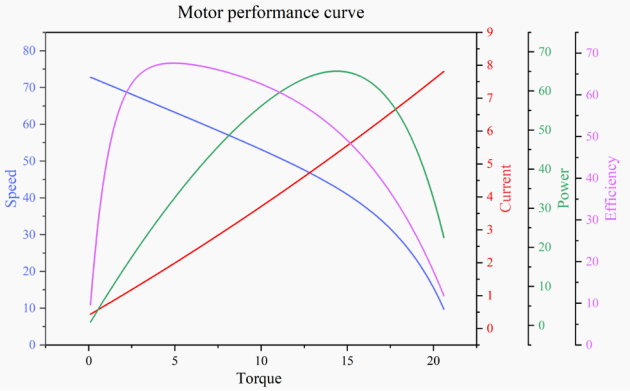
模组电气性能参数 Electrical Performance Parameters of the Module

电压范围 Voltage Range(V)	12-50.4	通讯协议 Communication	CANFD/CAN
峰值电流 Peak current(A)	10	扭矩控制精度 control accuracy(Nm/A)	±10%
转速控制精度 Speed control accuracy(Nm/A)	±10%	响应时间 Response time(us)	140
输入端编码器分辨率 High-speedend encoder resolution(bit)	14	输出端编码器分辨率 Low-speedend encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5	控制频率 Control frequency(Hz)	3K

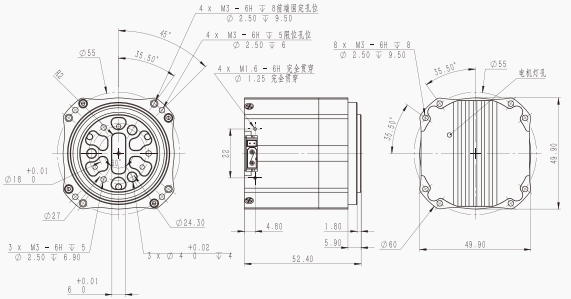
电机特性参数 Motor parameters

扭矩常数 Torque constant(Nm/A)	0.54	极对数 Polar logarithm	14

TN曲线 TN curve

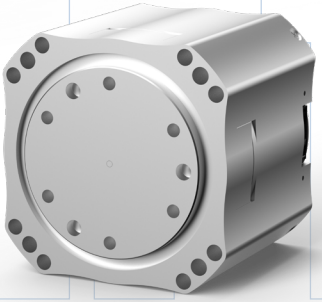


产品安装尺寸图纸 Product Installation Dimension Drawing



高擎 HIGH TORQUE

High Torque



HTDW-6036-02-DNE

行星关节模组

产品参数 Product Parameters

基本参数 Basic parameters

长宽 length and width (mm)	60
高度 Height (mm)	56
重量 weight (g)	565

减速机特性参数 Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

模组性能参数 Module Performance Parameters

减速比 Reduction ratio	36	堵转扭矩 Locked-rotor torque(Nm)	36
额定电压 Rated Voltage (V)	24	额定转速 Rated RPM(RPM)	50
额定电流 Rated Current (A)	3	空载转速 No-load RPM(RPM)	60
额定扭矩 Rated torque(Nm)	10	额定输出功率 Rated Output Power(W)	52

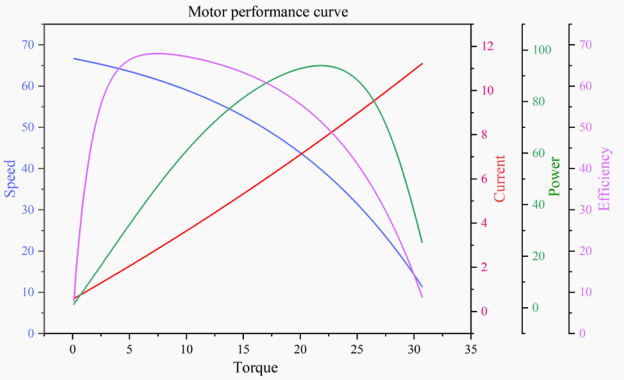
模组电气性能参数 Electrical Performance Parameters of the Module

电压范围 Voltage Range(V)	12-50.4	通讯协议 Communication	CANFD/CAN
峰值电流 Peak current(A)	15	扭矩控制精度 control accuracy(Nm/A)	±10%
转速控制精度 Speed control accuracy(Nm/A)	±10%	响应时间 Response time(us)	140
输入端编码器分辨率 High-speedend encoder resolution(bit)	14	输出端编码器分辨率 Low-speedend encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5	控制频率 Control frequency(Hz)	3K

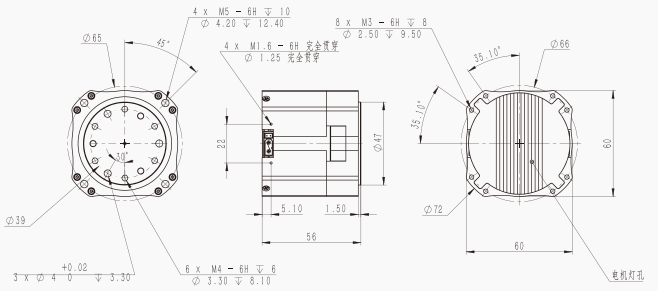
电机特性参数 Motor parameters

扭矩常数 Torque constant(Nm/A)	0.531	极对数 Polar logarithm	14

TN曲线 TN curve

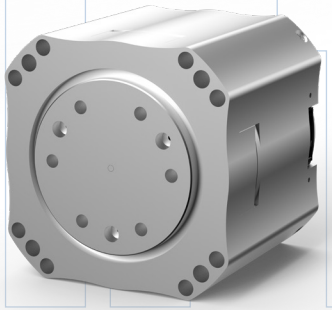


产品安装尺寸图纸 Product Installation Dimension Drawing



高擎 HIGH TORQUE

High Torque



HTDW-6036-02-CNE

（交叉滚子轴承版本） 行星关节模组

产品参数 Product Parameters

基本参数 Basic parameters

长宽 length and width（mm）	60
高度 Height（mm）	56
重量 weight（g）	570

减速机特性参数 Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

模组性能参数 Module Performance Parameters

减速比 Reduction ratio	36	堵转扭矩 Locked-rotor torque(Nm)	36
额定电压 Rated Voltage（V）	24	额定转速 Rated RPM(RPM)	50
额定电流 Rated Current（A）	3	空载转速 No-load RPM(RPM)	60
额定扭矩 Rated torque(Nm)	10	额定输出功率 Rated Output Power(W)	52

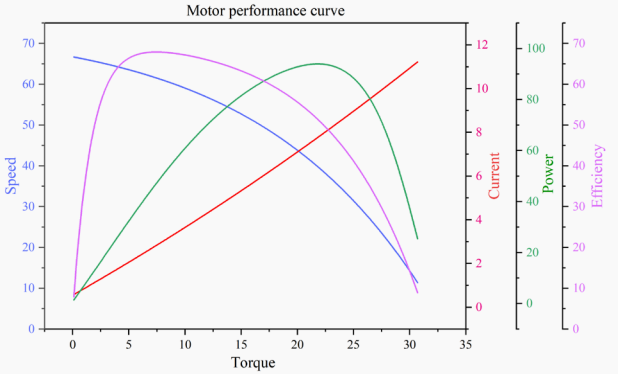
模组电气性能参数 Electrical Performance Parameters of the Module

电压范围 Voltage Range(V)	12-50.4	通讯协议 Communication	CANFD/CAN
峰值电流 Peak current(A)	15	扭矩控制精度 control accuracy(Nm/A)	±10%
转速控制精度 Speed control accuracy(Nm/A)	±10%	响应时间 Response time(us)	140
输入端编码器分辨率 High-speedend encoder resolution(bit)	14	输出端编码器分辨率 Low-speedend encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5	控制频率 Control frequency(Hz)	3K

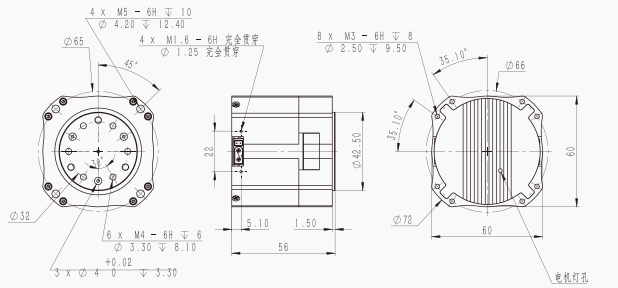
电机特性参数 Motor parameters

扭矩常数 Torque constant(Nm/A)	0.531	极对数 Polar logarithm	14

TN曲线 TN curve

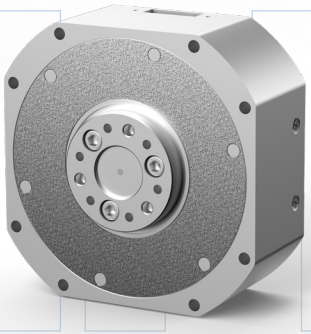


产品安装尺寸图纸 Product Installation Dimension Drawing



高擎 HIGH TORQUE

High Torque



HTPW-7507-02-DNE

行星关节模组

产品参数 Product Parameters

基本参数 Basic parameters

长宽 length and width (mm)	75
高度 Height (mm)	35
重量 weight (g)	390

减速机特性参数 Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

模组性能参数 Module Performance Parameters

减速比 Reduction ratio	7	堵转扭矩 Locked-rotor torque(Nm)	12
额定电压 Rated Voltage (V)	24	额定转速 Rated RPM(RPM)	250
额定电流 Rated Current (A)	5.6	空载转速 No-load RPM(RPM)	270
额定扭矩 Rated torque(Nm)	4	额定输出功率 Rated Output Power(W)	92

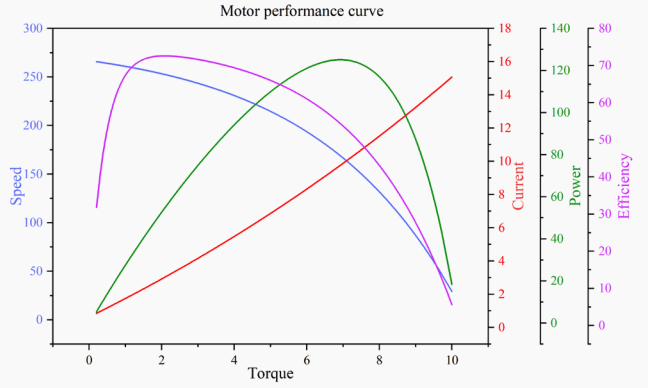
模组电气性能参数 Electrical Performance Parameters of the Module

电压范围 Voltage Range(V)	12-50.4	通讯协议 Communication	CANFD/CAN
峰值电流 Peak current(A)	20	扭矩控制精度 control accuracy(Nm/A)	±10%
转速控制精度 Speed control accuracy(Nm/A)	±10%	响应时间 Response time(us)	140
输入端编码器分辨率 High-speedend encoder resolution(bit)	14	输出端编码器分辨率 Low-speedend encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5	控制频率 Control frequency(Hz)	3K

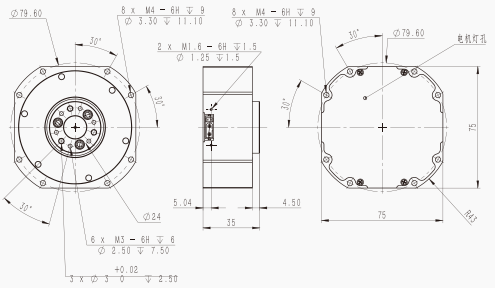
电机特性参数 Motor parameters

扭矩常数 Torque constant(Nm/A)	0.602	极对数 Polar logarithm	14

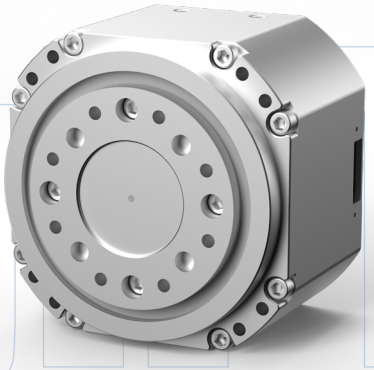
TN曲线 TN curve



产品安装尺寸图纸 Product Installation Dimension Drawing



高擎 HIGH TORQUE



HTDW-7535-02-CNE

（交叉滚子轴承版本） 行星关节模组

产品参数 Product Parameters

基本参数 Basic parameters

长宽 length and width（mm）	75
高度 Height（mm）	56
重量 weight（g）	850

减速机特性参数 Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

模组性能参数 Module Performance Parameters

减速比 Reduction ratio	35	堵转扭矩 Locked-rotor torque(Nm)	60
额定电压 Rated Voltage（V）	24	额定转速 Rated RPM(RPM)	50
额定电流 Rated Current（A）	4.2	空载转速 No-load RPM(RPM)	60
额定扭矩 Rated torque(Nm)	18	额定输出功率 Rated Output Power(W)	94

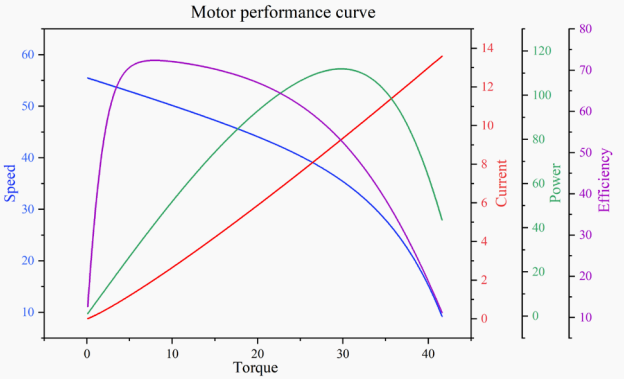
模组电气性能参数 Electrical Performance Parameters of the Module

电压范围 Voltage Range(V)	12-50.4	通讯协议 Communication	CANFD/CAN
峰值电流 Peak current(A)	20	扭矩控制精度 control accuracy(Nm/A)	±10%
转速控制精度 Speed control accuracy(Nm/A)	±10%	响应时间 Response time(us)	140
输入端编码器分辨率 High-speedend encoder resolution(bit)	14	输出端编码器分辨率 Low-speedend encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5	控制频率 Control frequency(Hz)	3K

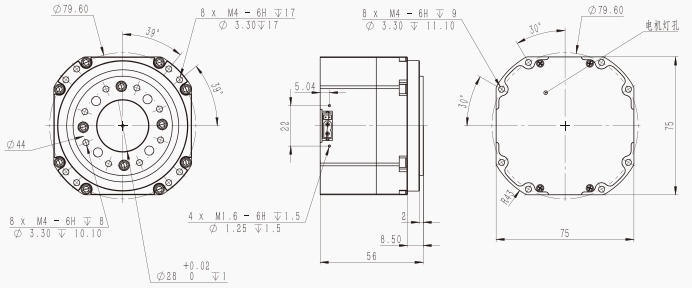
电机特性参数 Motor parameters

扭矩常数 Torque constant(Nm/A)	0.602	极对数 Polar logarithm	14

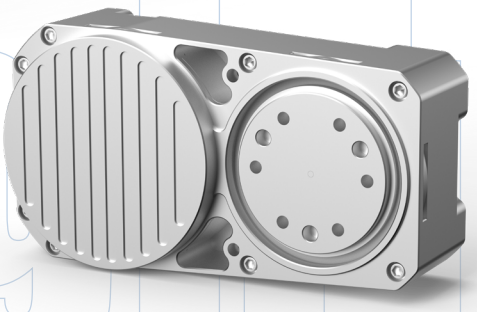
TN曲线 TN curve



产品安装尺寸图纸 Product Installation Dimension Drawing



高擎 HIGH TORQUE



HTPU-6035-04-CNE

超薄系列关节模组

产品参数 Product Parameters

基本参数 Basic parameters

长宽 length and width (mm)	115X61.9
高度 Height (mm)	37
重量 weight (g)	558

减速机特性参数 Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

模组性能参数 Module Performance Parameters

减速比 Reduction ratio	35	堵转扭矩 Locked-rotor torque(Nm)	36
额定电压 Rated Voltage (V)	24	额定转速 Rated RPM(RPM)	50
额定电流 Rated Current (A)	3	空载转速 No-load RPM(RPM)	60
额定扭矩 Rated torque(Nm)	10	额定输出功率 Rated Output Power(W)	52

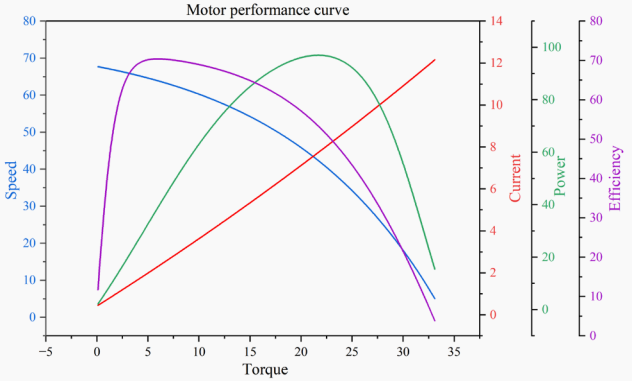
模组电气性能参数 Electrical Performance Parameters of the Module

电压范围 Voltage Range(V)	12-50.4	通讯协议 Communication	CANFD/CAN
峰值电流 Peak current(A)	15	扭矩控制精度 control accuracy(Nm/A)	±10%
转速控制精度 Speed control accuracy(Nm/A)	±10%	响应时间 Response time(us)	140
输入端编码器分辨率 High-speedend encoder resolution(bit)	14	输出端编码器分辨率 Low-speedend encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5	控制频率 Control frequency(Hz)	3K

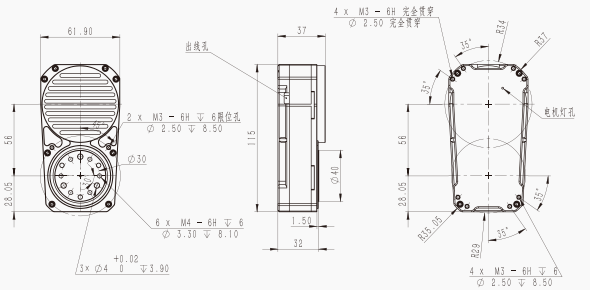
电机特性参数 Motor parameters

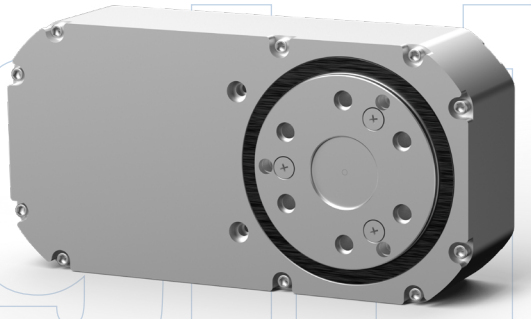
扭矩常数 Torque constant(Nm/A)	0.531	极对数 Polar logarithm	14

TN曲线 TN curve



产品安装尺寸图纸 Product Installation Dimension Drawing





HTPU-7033-04-CNE

超薄系列关节模组

产品参数 Product Parameters

基本参数 Basic parameters

长宽 length and width (mm)	140.5X71.5
高度 Height (mm)	36.9
重量 weight (g)	808

减速机特性参数 Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

模组性能参数 Module Performance Parameters

减速比 Reduction ratio	33	堵转扭矩 Locked-rotor torque(Nm)	50
额定电压 Rated Voltage (V)	24	额定转速 Rated RPM(RPM)	40
额定电流 Rated Current (A)	4.8	空载转速 No-load RPM(RPM)	48
额定扭矩 Rated torque(Nm)	20	额定输出功率 Rated Output Power(W)	84

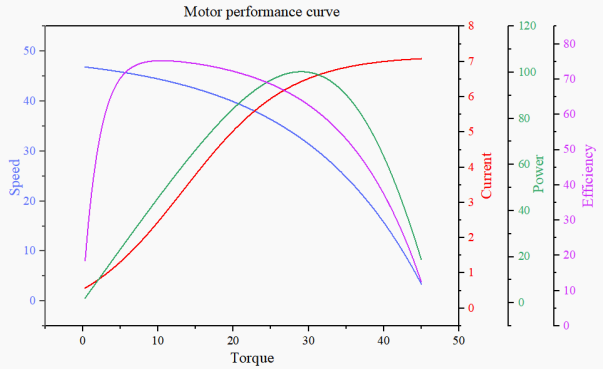
模组电气性能参数 Electrical Performance Parameters of the Module

电压范围 Voltage Range(V)	12-50.4	通讯协议 Communication	CANFD/CAN
峰值电流 Peak current(A)	20	扭矩控制精度 control accuracy(Nm/A)	±10%
转速控制精度 Speed control accuracy(Nm/A)	±10%	响应时间 Response time(us)	140
输入端编码器分辨率 High-speedend encoder resolution(bit)	14	输出端编码器分辨率 Low-speedend encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5	控制频率 Control frequency(Hz)	3K

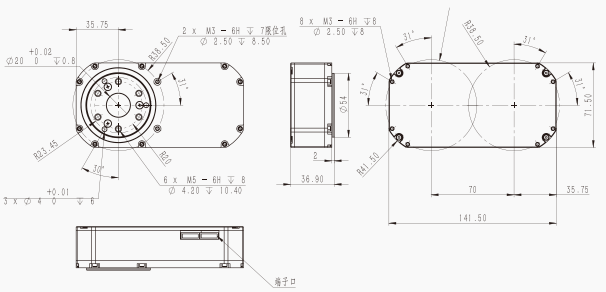
电机特性参数 Motor parameters

扭矩常数 Torque constant(Nm/A)	0.59	极对数 Polar logarithm	14

TN曲线 TN curve

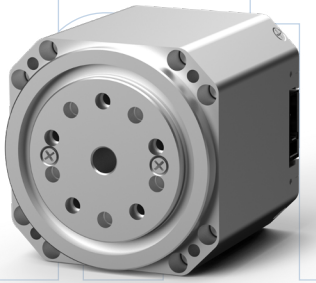


产品安装尺寸图纸 Product Installation Dimension Drawing



高擎 HIGH TORQUE

High Torque



HTCP-4531-06-CYC

中空关节模组

产品参数 Product Parameters

基本参数 Basic parameters

长宽 length and width (mm)	44
高度 Height (mm)	44.9
重量 weight (g)	237

减速机特性参数 Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

模组性能参数 Module Performance Parameters

减速比 Reduction ratio	31	堵转扭矩 Locked-rotor torque(Nm)	10
额定电压 Rated Voltage (V)	24	额定转速 Rated RPM(RPM)	40
额定电流 Rated Current (A)	0.7	空载转速 No-load RPM(RPM)	160
额定扭矩 Rated torque(Nm)	2	额定输出功率 Rated Output Power(W)	8.5

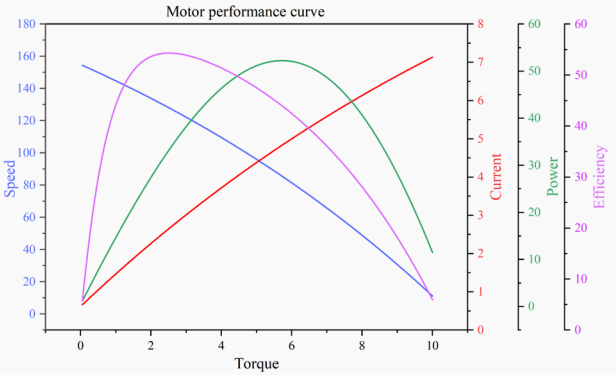
模组电气性能参数 Electrical Performance Parameters of the Module

电压范围 Voltage Range(V)	12-50.4	通讯协议 Communication	CANFD/CAN
峰值电流 Peak current(A)	10	扭矩控制精度 control accuracy(Nm/A)	±10%
转速控制精度 Speed control accuracy(Nm/A)	±10%	响应时间 Response time(us)	140
输入端编码器分辨率 High-speedend encoder resolution(bit)	14	输出端编码器分辨率 Low-speedend encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5	控制频率 Control frequency(Hz)	3K

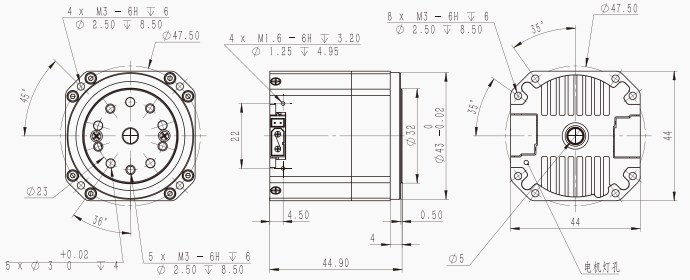
电机特性参数 Motor parameters

扭矩常数 Torque constant(Nm/A)	0.272	极对数 Polar logarithm	14

TN曲线 TN curve

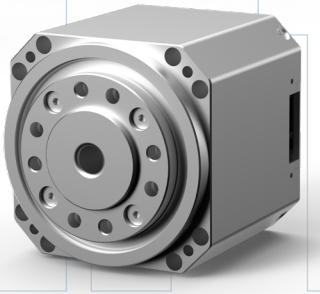


产品安装尺寸图纸 Product Installation Dimension Drawing



高擎 HIGH TORQUE

High Torque



HTCP-5031-06-CYC

中空关节模组

产品参数 Product Parameters

基本参数 Basic parameters

长宽 length and width (mm)	50
高度 Height (mm)	53
重量 weight (g)	385

减速机特性参数 Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

模组性能参数 Module Performance Parameters

减速比 Reduction ratio	31	堵转扭矩 Locked-rotor torque(Nm)	16
额定电压 Rated Voltage (V)	24	额定转速 Rated RPM(RPM)	50
额定电流 Rated Current (A)	2	空载转速 No-load RPM(RPM)	103
额定扭矩 Rated torque(Nm)	4	额定输出功率 Rated Output Power(W)	21

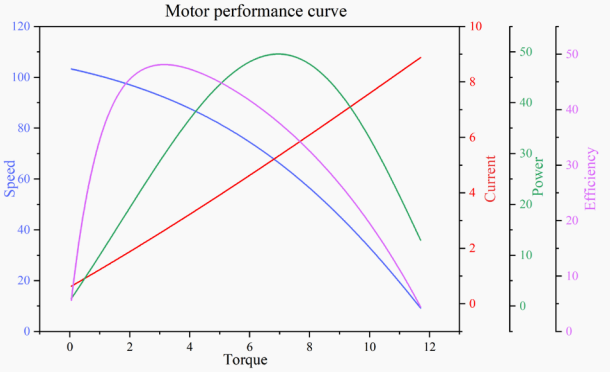
模组电气性能参数 Electrical Performance Parameters of the Module

电压范围 Voltage Range(V)	12-50.4	通讯协议 Communication	CANFD/CAN
峰值电流 Peak current(A)	10	扭矩控制精度 control accuracy(Nm/A)	±10%
转速控制精度 Speed control accuracy(Nm/A)	±10%	响应时间 Response time(us)	140
输入端编码器分辨率 High-speedend encoder resolution(bit)	14	输出端编码器分辨率 Low-speedend encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5	控制频率 Control frequency(Hz)	3K

电机特性参数 Motor parameters

扭矩常数 Torque constant(Nm/A)	0.54	极对数 Polar logarithm	14

TN曲线 TN curve



产品安装尺寸图纸 Product Installation Dimension Drawing

